

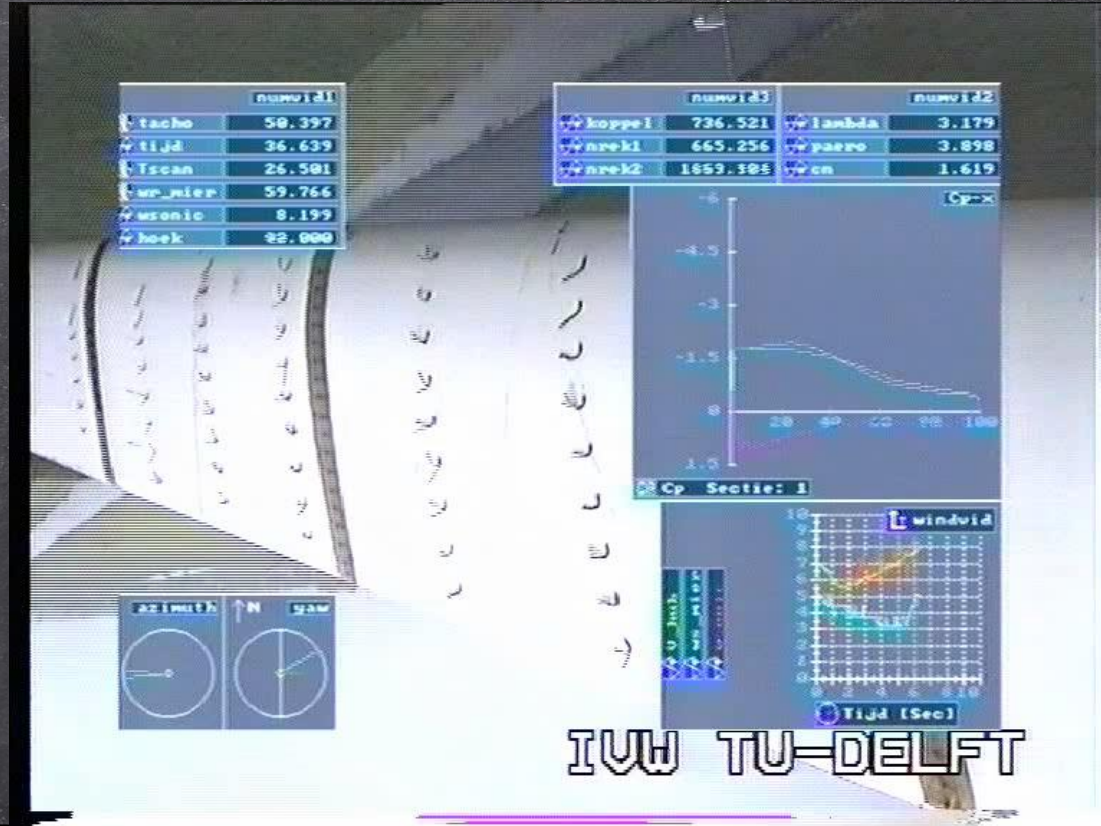
Rotor and Wake Aerodynamics

Introduction to unsteady aerodynamics

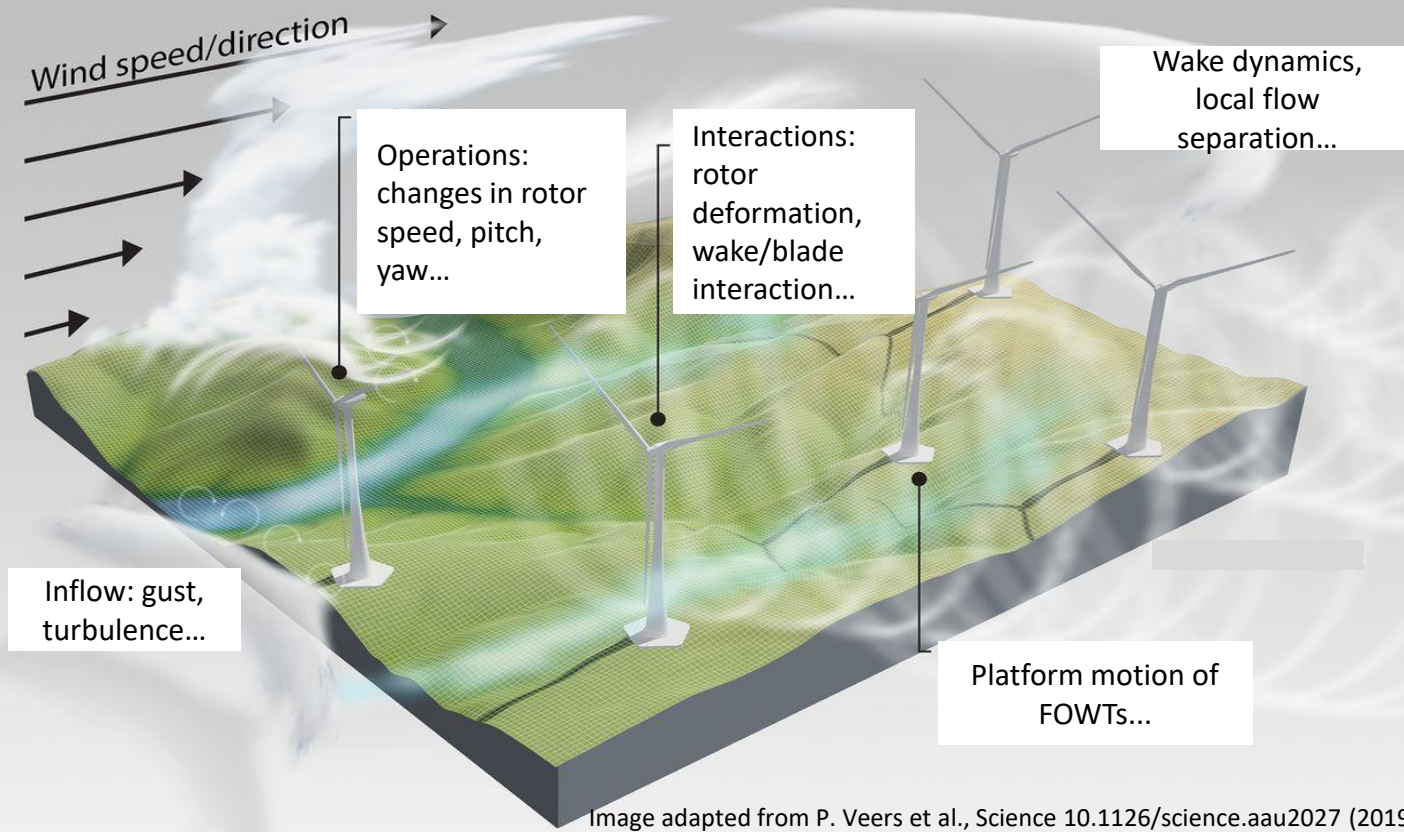
Learning objectives

- To define sources of flow unsteadiness
- To evaluate the level of flow unsteadiness

Measurement of unsteady effects on a rotor



Sources of flow unsteadiness



Types of unsteadiness:
periodic,
Aperiodic
and
transient

When do we call a flow unsteady?

- All flows are unsteady, the flow varies in time
- However, in many cases, the unsteady effects are small for our practical purposes
- We can say that A flow is unsteady when a "steady flow assumption" leads to a significant error

Characterisation of flow unsteadiness



Example	U_0 [m/s]	Frequency [Hz]	Chord length [m]	Unsteady level
Mosquito	2	600	0.002	?
Seagull (flapping)	15	3	0.2	?
Wind turbine blade in gust	10	0.2	1	?

Characterisation of flow unsteadiness

$$k = \frac{\omega c}{2V}$$

Reduced frequency - determines the degree of unsteadiness of the problem.
 ω is the angular frequency of the unsteady behavior.

- for $k=0$, the flow is steady;
- for $k < 0.05$ the system can be considered quasi-steady;
- for $0.05 < k < 0.2$ is unsteady
- for $0.2 < k$ is highly unsteady (dominated by acceleration terms)

Note: ω in [rad/s]

$$s = \frac{2}{c} \int_0^t V dt$$

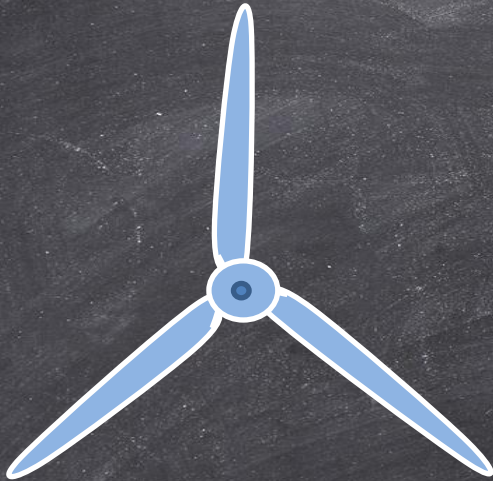
Reduced time - determines the relative distance of traveling of the airfoil in relation to its chord

Characterisation of flow unsteadiness

	Example	U ₀ [m/s]	Frequency [Hz]	Chord length [m]	Reduced frequency k [-]	Unsteady level
	Mosquito	2	600	0.002	1.88	Highly unsteady
	Seagull (flapping)	15	3	0.2	0.13	Unsteady
	Wind turbine blade in gust	10	0.1	1	0.03	Quasi- steady

$$k = \frac{\omega C}{2V}$$

Characterisation of flow unsteadiness



Property	lift @ tip	lift @ root	rotor thrust
k	0.03	0.12	3.14
Unsteady level	Quasi-steady	Unsteady	Highly unsteady

$D = 100\text{m}$, chord (tip) = 1m , chord (root) = 4m , mean incoming flow speed $V_0 = 10\text{m/s}$, under a gust with frequency of 0.1 Hz .

$$k = \frac{\omega D}{V} \quad \text{for rotor}$$

In this module we have

- Discussed the sources of flow unsteadiness;
- Learnt how to evaluate the flow unsteadiness.

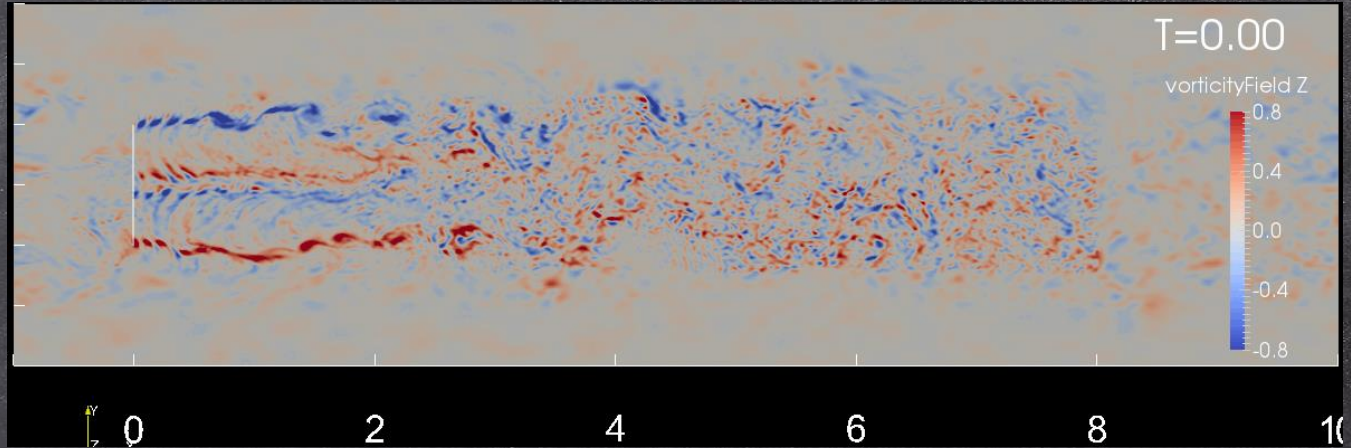
Rotor and Wake Aerodynamics

Unsteady rotor aerodynamics

Learning objectives

- To understand the unsteady rotor aerodynamics
- To learn about basic applications of models for unsteady rotor flow
- To learn a few dynamic inflow models and its application

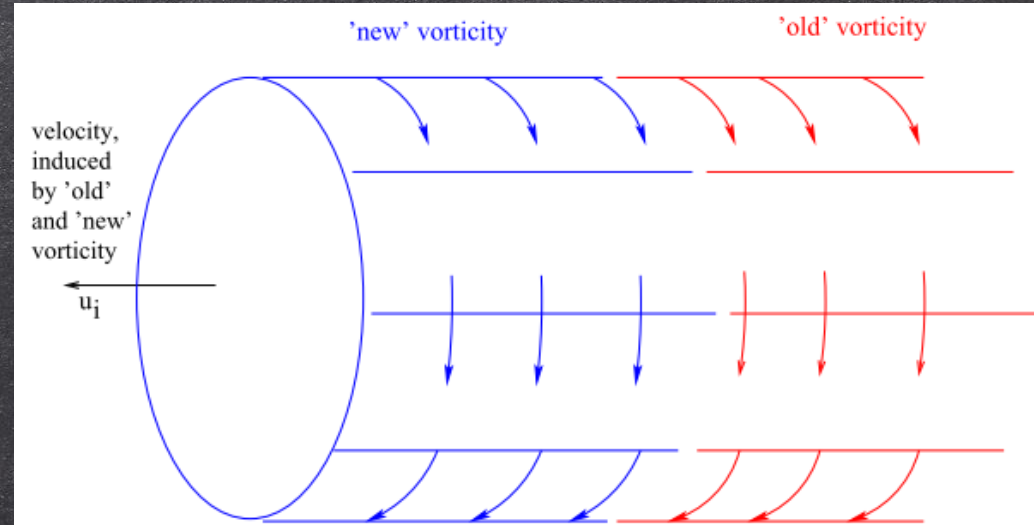
Dynamic inflow (induction)



Dynamic inflow : time lagging in the response of the induced velocity field of a rotor following rapid changes in the rotor operating state.
The characteristic time scale: D/V_0 .

Rotor quasi-steady response

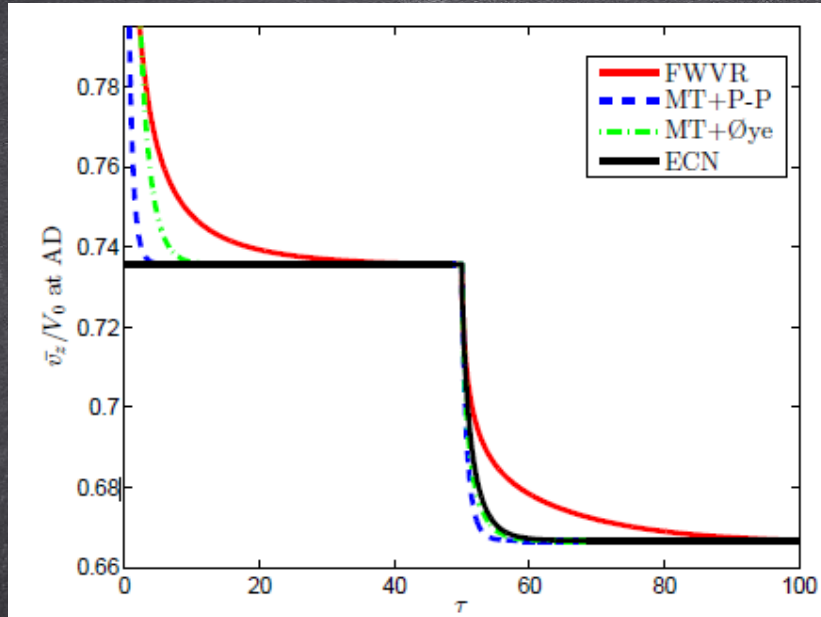
Change in the load



Wake with mixed vorticity during a step change in load, e.g. pitch angle step.

Schepers, G. "Engineering models in wind energy aerodynamics" (2012).

Rotor indicial response



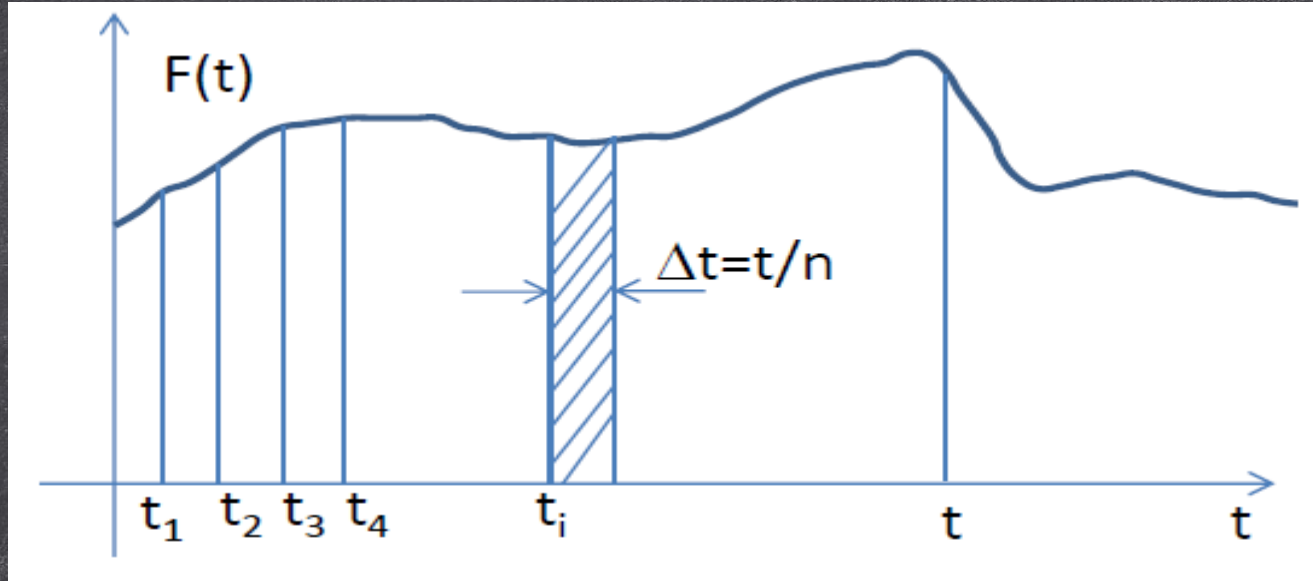
Indicial response of the axial velocity at AD:

$$\Phi_d(r_j, t) = 1 - \beta_j e^{\omega_{1j} t} - (1 - \beta_j) e^{\omega_{2j} t}$$

Comparison of averaged axial velocity at the actuator disc between the FWVR model, the Pitt-Peters, the $\emptyset$ ye and ECN model for C_t 7/9 $\rightarrow$ 8/9.

Yu, W., The wake of an unsteady actuator disc, Delft University of Technology, 2018

Duhamel's Integral and its differential form



Arbitrary excitation force, $F(t)$, split into n impulses

Duhamel's Integral and its differential form

The induced velocity output at the AD $u_d(r, t)$:

$$u_d(r, t) = \gamma(0)\Phi_d(r, t) + \int_0^t \frac{d\gamma}{d\sigma} \Phi_d(r, t - \sigma) d\sigma$$

σ is a dummy time variable for integration

Duhamel's Integral and its differential form

Substitute indicial response into the Duhamel's integral :

$$c_{k,j}(t) = \int_0^t \gamma'(\sigma) A_{k,j} e^{\omega_{k,j}(t-\sigma)} d\sigma$$

Differentiating it :

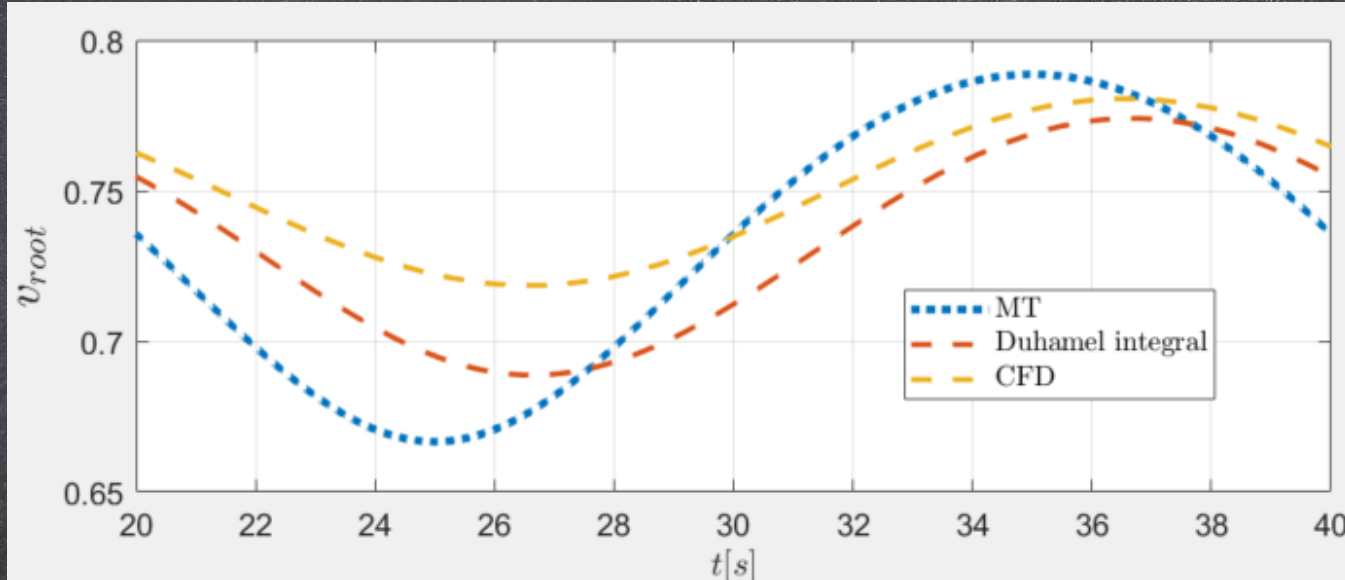
$$\dot{c}_{k,j}(t) - \omega_{k,j} c_{k,j}(t) = A_{k,j} \dot{\gamma}(t), k = 1, 2$$

The final rotor velocity :

$$u_d(r, t) = \gamma(t) - c_{1,j}(t) - c_{2,j}(t)$$

Dynamic rotor response

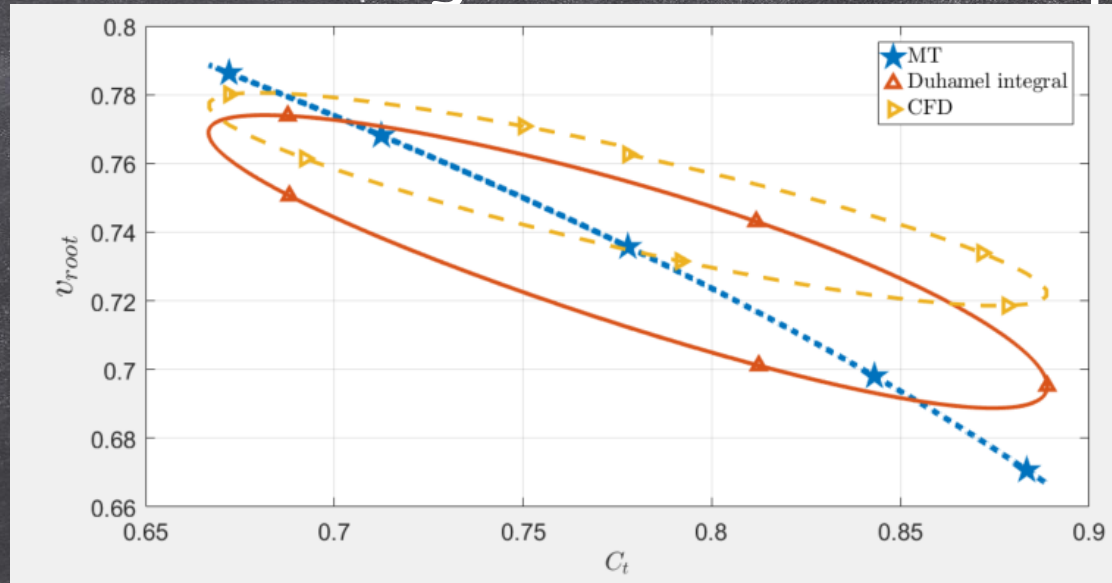
--- Duhamel integral of indicial response



The axial velocity at $r/R=0$ from the steady MT, Duhamel integral and CFD for an AD undergoing harmonic thrust oscillations for baseline $C_t=7/9$ with an amplitude of $\Delta C_t=1/9$, for $k=0.05$.

Dynamic rotor response

--- Duhamel integral of indicial response



Hysteresis loops of the axial velocity at $r/R=0$ between MT, Duhamel integral, CFD for an AD undergoing harmonic thrust oscillations for baseline $C_t = 7/9$ with an amplitude of $\Delta C_t = 1/9$, for $k = 0.05$.

The Pitt-Peters dynamic inflow model

The linear Pitt-Peters dynamic-inflow model is shown in the equation below:

- for an annulus at radius r_j with area A_j
- for a loading on the annulus C_{tj}
- where v_j is the perturbation velocity at the rotor (induced velocity)

$$\frac{1}{\rho A_j V_0^2 / 2} \left[\frac{8}{3\pi} \rho A_j r_j \frac{dv_j}{dt} + 2\rho A_j v_j (V_0 + v_j) \right] = C_{tj}$$

The Øye dynamic inflow model

The Øye dynamic-inflow model differs from the Pitt-Peters model. Instead of expressing a linear form of the momentum equation, it expresses a filtering of the quasi-steady values through two first-order differential equations. It's applicable to annulus.

$$v_{int} + \tau_1 \frac{dv_{int}}{dt} = v_{qs} + b\tau_1 \frac{dv_{qs}}{dt}.$$

$$v_z + \tau_2 \frac{dv_z}{dt} = v_{int}.$$

$$\tau_1 = \frac{1.1}{(1 - 1.3a)} \frac{R}{V_0}.$$

$$\tau_2 = (0.39 - 0.26(\frac{r_j}{R})^2)\tau_1.$$

b: 0.6; v_{qs} : quasi-steady induced velocity from BEM; v_{int} : intermediate induced velocity; v_z : the final induced velocity, a is the axial induction factor.

The Larsen-Madsen dynamic inflow model

In this model, as in Øye's model, dynamic inflow is modelled using a low pass filtering of the steady state induced velocities.

$$a_{t_{i+1}} = a_{t_i} e^{\frac{-\Delta t}{\tau}} + a_{qs} \left(1 - e^{\frac{-\Delta t}{\tau}}\right)$$

$$\tau = 0.5 \frac{R}{U_{wake}} \simeq 0.5 \frac{R}{aU_{\infty}}$$

a_{qs} : quasi-steady induced velocity at time step of $i+1$; U_{wake} : wake convection velocity; a : induction factor at current step

In this module we have

- Learnt how basic models can be used to quantify the unsteady rotor flow response
- Learnt a few engineering dynamic inflow models

Rotor and Wake Aerodynamics

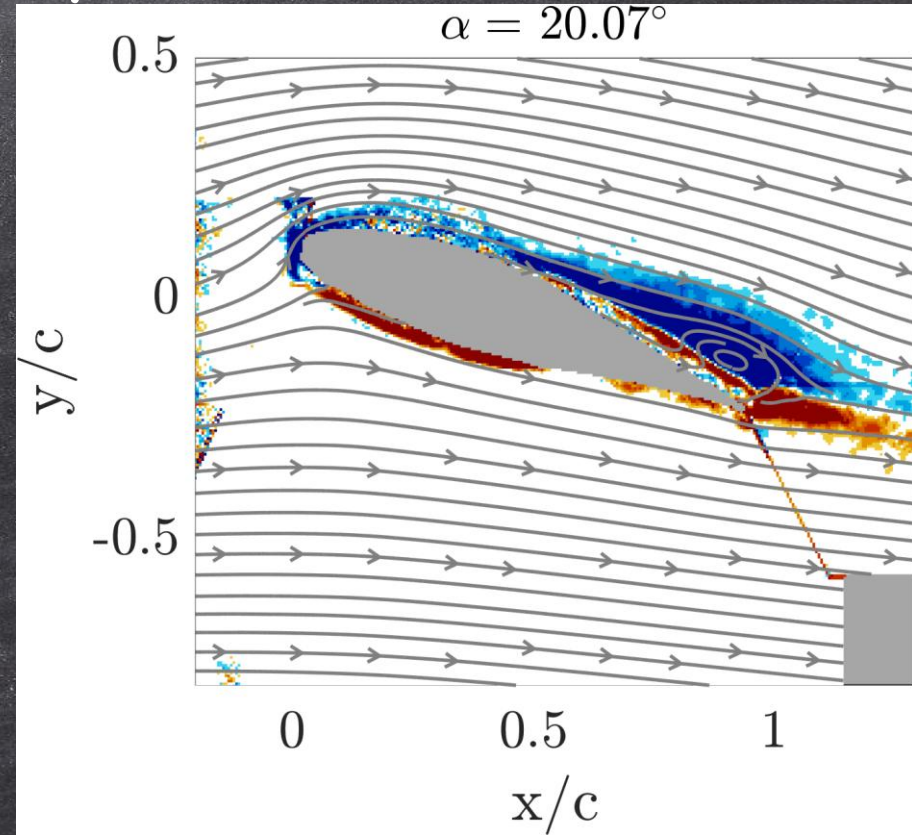
Unsteady airfoil aerodynamics

Learning objectives

- To understand the unsteady airfoil aerodynamics
- To understand the main physics of dynamic stall
- To implement a dynamic stall model in BEM

Unsteady airfoil aerodynamics

Unsteady airfoil aerodynamics (Dynamic stall) : time lagging in the response of the induced velocity field of an airfoil following rapid changes in operating conditions.
The characteristic time scale: c/V_0 .



Airfoil indicial response (Wagner's function)

Wagner obtained the lift coefficient on a thin-airfoil undergoing a step change in AoA in incompressible flow (indicial lift response):

$$C_l(t) = \underbrace{\frac{\pi c}{2U_\infty} \delta(t)}_{\text{Non-circulatory effects}} + \underbrace{2\pi\alpha \phi(s)}_{\text{Circulatory effects}}$$

Non-circulatory effects : fluid acceleration
(a.k.a “added mass”, “apparent mass”)

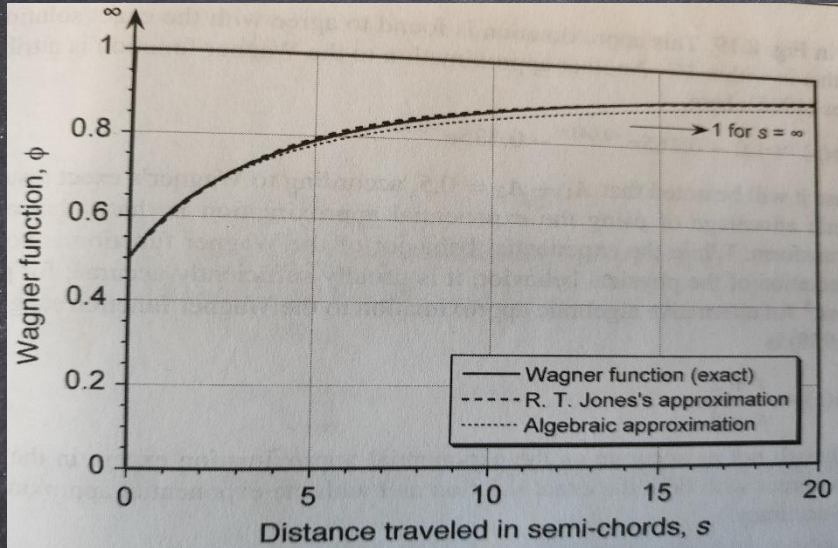
Circulatory effects: bound vorticity or
circulation on the airfoil, governed by
Kutta-Joukowski theorem

$\delta(t)$ is the Dirac delta

$\phi(s)$ is the Wagner's function

Leishman, G., Principles of helicopter aerodynamics
Cambridge Aerospace Series, 2008

Airfoil indicial response (Wagner's function)



Jone's numerical approximation:

$$\phi(s) = 1 - 0,165e^{-0,045s} - 0,335e^{-0,3s}$$

Wagner's function for a step change in AoA

Leishman, G., Principles of helicopter aerodynamics, Cambridge Aerospace Series, 2008

Quasi-steady response of an airfoil

We will consider the effect of the perceived velocity at the body

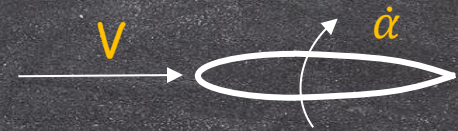
We will first use quasi-steady theory

Quasi-Steady Thin Airfoil Theory

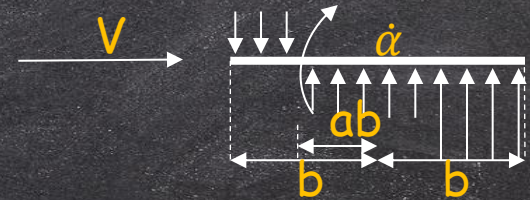
Plunge motion

Pitch motion

Airfoil motion



Flatplate motion



Perceived windspeed

$$[V; w = -\dot{h}]$$

$$w = \dot{\alpha}(x - ab)$$

Equivalent angle of attack

$$\alpha = \frac{-\dot{h}}{V}$$

$$\alpha = b \left(\frac{1}{2} - a \right) \frac{\dot{\alpha}}{V}$$

Quasi-Steady Thin Airfoil Theory

Quasi-steady

Translation

Plunge motion

Pitch motion



$$C_l = 2\pi \left[\alpha + \frac{-\dot{h}}{V} + b \left(\frac{1}{2} - a \right) \frac{\dot{\alpha}}{V} \right]$$

Note: if the pitching axis is at $1/4c$, then $a = -1/2$.

$$C_l = 2\pi\alpha_{qs} \quad \alpha_{qs} = \alpha + \frac{-\dot{h}}{V} + c \frac{\dot{\alpha}}{V}$$

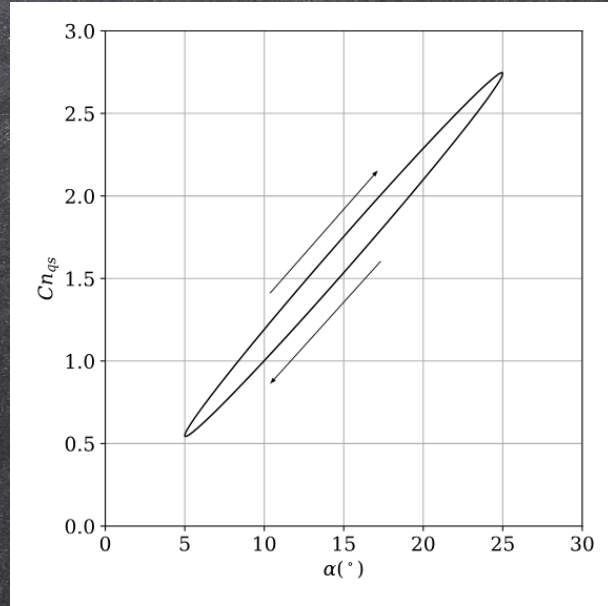
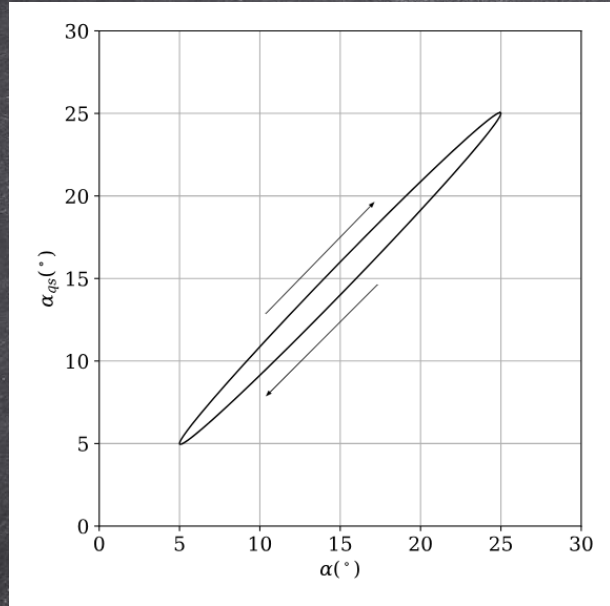
Assuming small AoA: $C_n \approx C_l$

Quasi-Steady Thin Airfoil Theory

For airfoils with camber, we introduce the angle for which C_n is null, named α_0 .

$$Cn_{qs} = 2\pi(\alpha_{qs} - \alpha_0)$$

Quasi-Steady Thin Airfoil Theory



Quasi-steady AoA (α_{qs}) and normal force (Cn_{qs}) for a **flat plate** pitching in a sinusoidal motion $\alpha = 5 \sin(\omega t) + 15$, with $\omega = k \cdot (2c)/U_{inf}$, with $k = 0.1$. The pitching axis $\frac{1}{4}$ * chord.

Quasi-Steady Thin Airfoil Theory

With the superposition of indicial aerodynamic responses, using the Duhamel integral, we obtain the lift response to an arbitrary change in AoA:

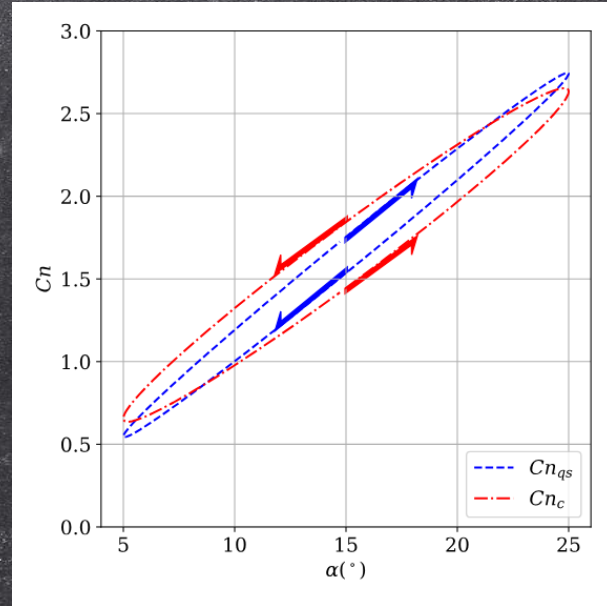
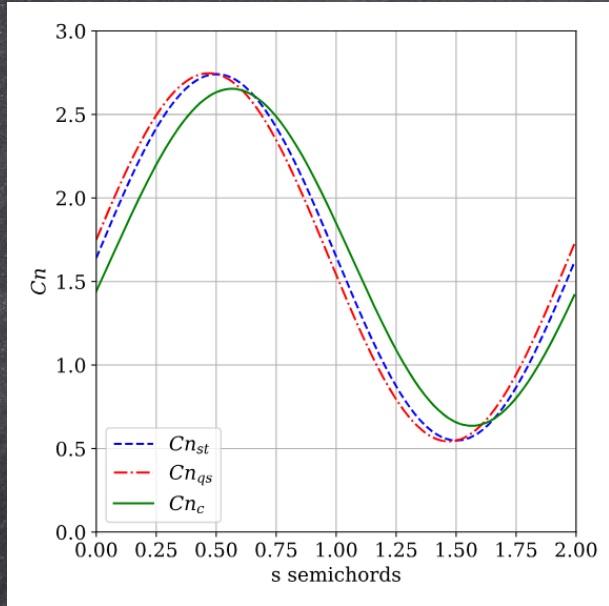
$$C_l(t) = \pi b \left(\frac{\dot{\alpha}}{U_\infty} + \frac{\ddot{h}}{U_\infty^2} - \frac{ba\ddot{\alpha}}{U_\infty^2} \right) + 2\pi \left(\alpha(0)\phi(s) + \int_0^s \frac{d\alpha(\sigma)}{dt} \phi(s-\sigma) d\sigma \right)$$

Similarly, using the differential term of the duhamel integral using the quasi-steady and indicial response function, we have

$$\begin{aligned}\alpha_{E(s)} &= \alpha_{qs(s)} - X_{lag(s)} - Y_{lag(s)} \\ X_{lag(s)} &= X_{lag(s-\Delta s)} e^{-b_1 \Delta s} + \Delta \alpha_{qs(s)} A_1 e^{-b_1 \frac{\Delta s}{2}} \\ Y_{lag(s)} &= Y_{lag(s-\Delta s)} e^{-b_2 \Delta s} + \Delta \alpha_{qs(s)} A_2 e^{-b_2 \frac{\Delta s}{2}}\end{aligned}$$

Dynamic airfoil response

--- Duhamel integral of indicial response



Comparison of the normal load coefficients for the steady flow $C_{n_{st}}$, quasi-steady $C_{n_{qs}}$ and the circulatory term of the unsteady normal load coefficient C_{n_c} . The effect of considering unsteady aerodynamics in phase delay and damping.

Dynamic airfoil response

--- Duhamel integral of indicial response

The non-circulatory normal force for a pitching airfoil in incompressible flows, is approximated by

$$C_{n_{nc}(t)} = \frac{4 * K_{\alpha} c}{U_{\infty}} \left(\frac{d\alpha_{qs}(t)}{dt} - D_{nc}(t) \right)$$

where $D_{nc}(t)$ is a deficiency function that can be solved by integrating in time as

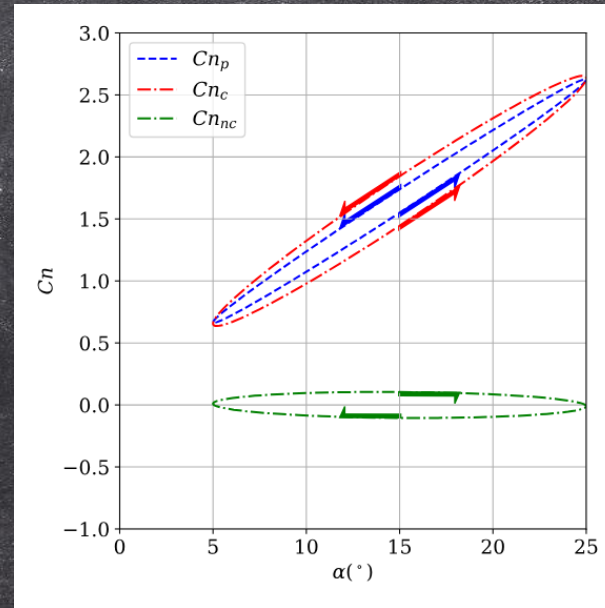
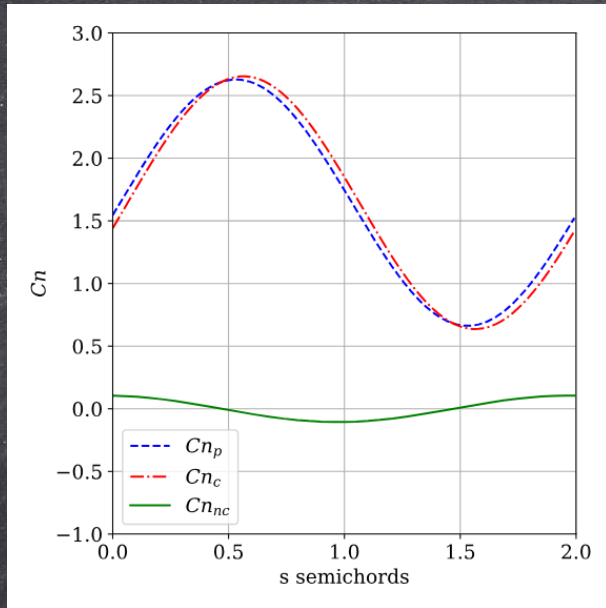
$$D_{nc}(t) = D_{nc}(t-\Delta t) e^{\frac{-a_{so} \Delta t}{K_{\alpha} c}} + \left(\frac{d\alpha_{qs}(t)}{dt} - \frac{d\alpha_{qs}(t-\Delta t)}{dt} \right) e^{\frac{-a_{so} \Delta t}{2K_{\alpha} c}}$$

The total unsteady potential-flow normal load coefficient is

$$C_{n_p} = C_{n_c} + C_{n_{nc}}$$

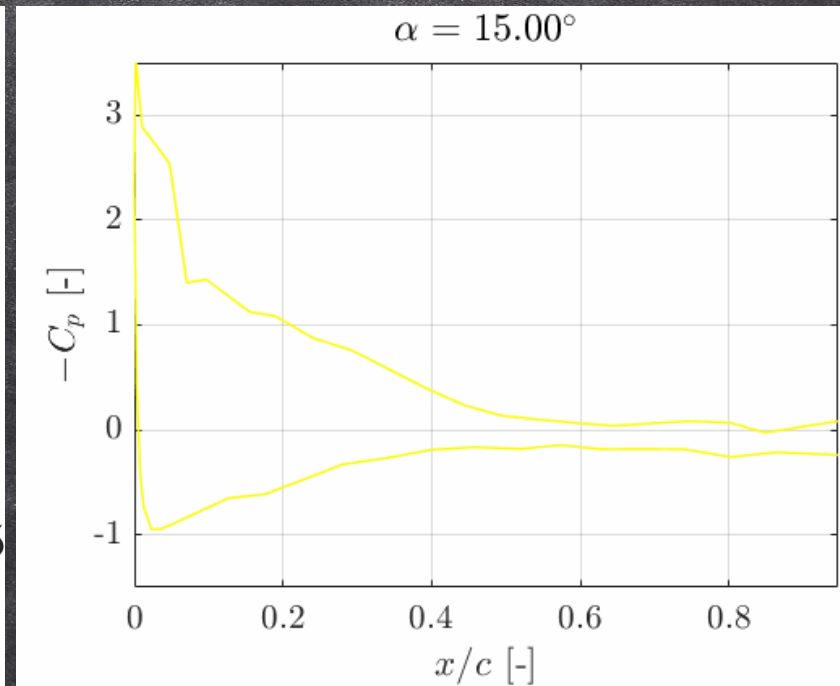
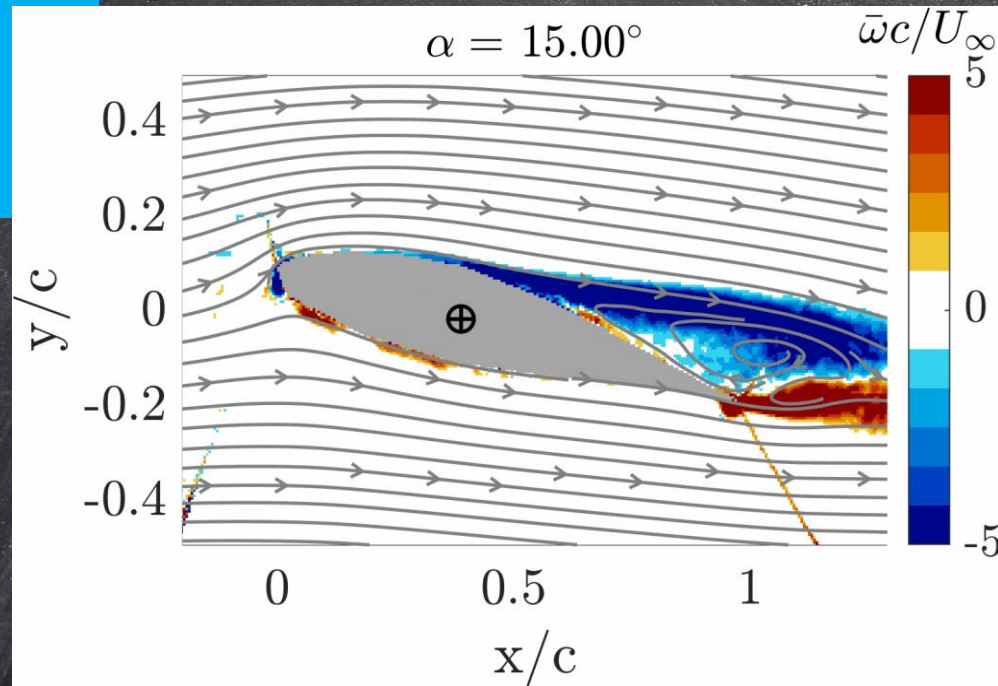
Dynamic airfoil response

--- Duhamel integral of indicial response



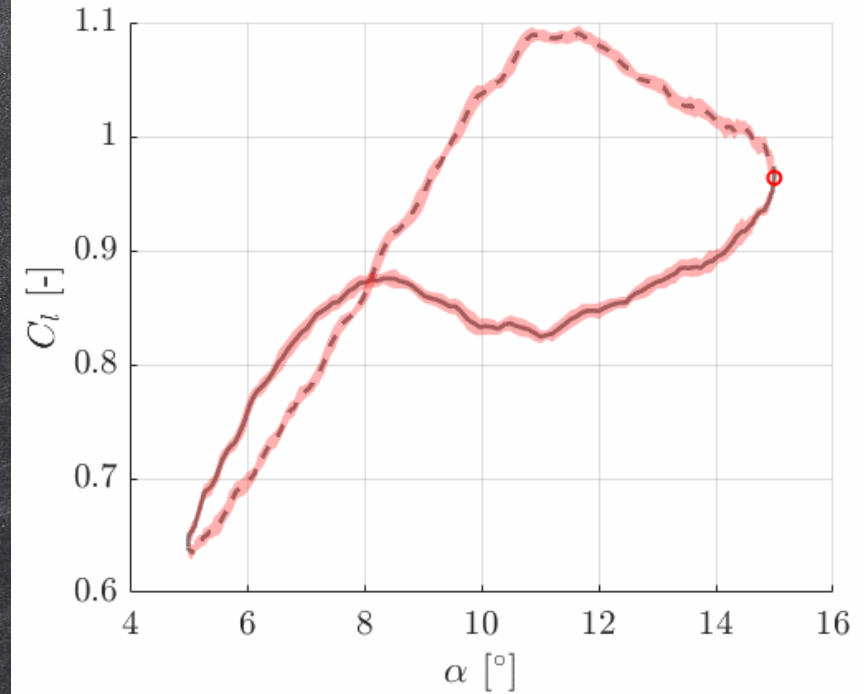
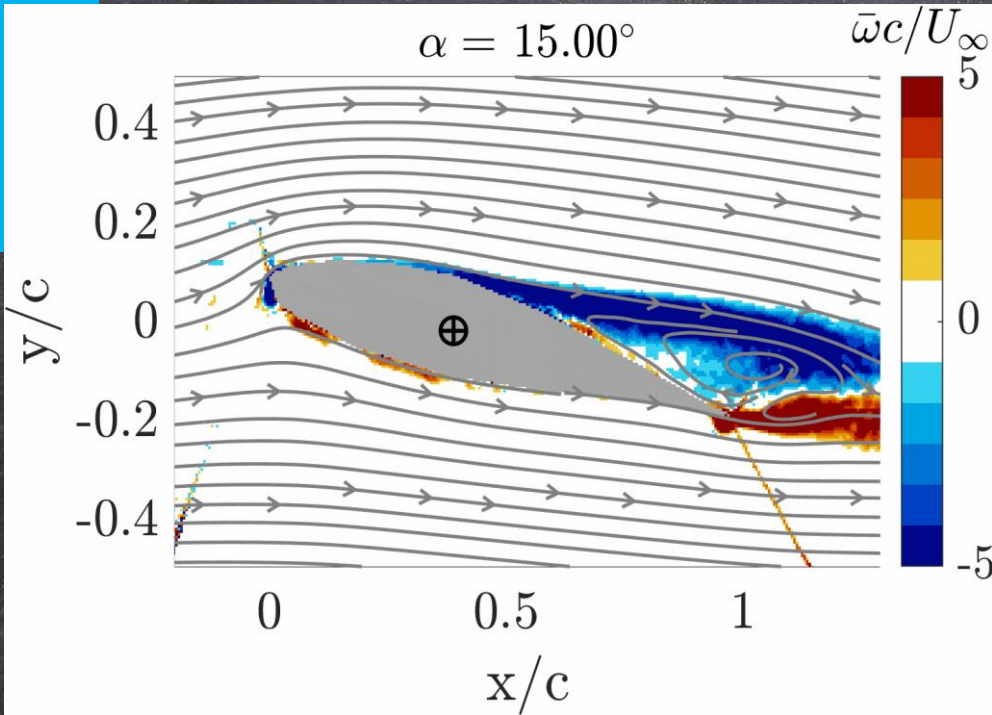
Comparison of the circulatory and non-circulatory terms of the unsteady normal load coefficient.

Introduction to dynamic stall



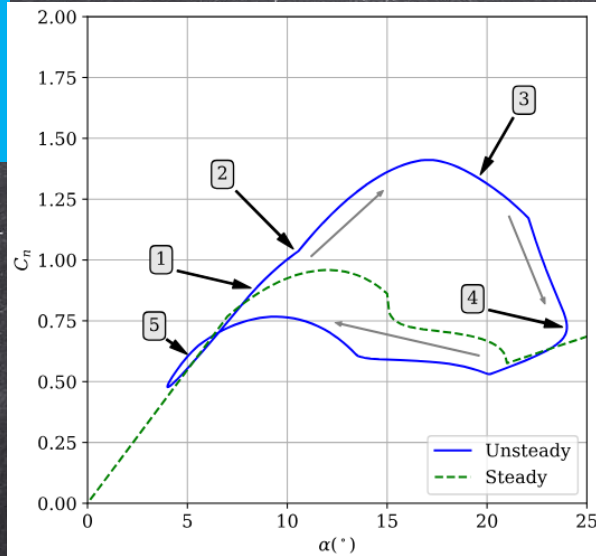
Case: $\bar{\alpha} = 10^\circ$, $k = 0.05$, $\Delta\alpha = 5^\circ$

Introduction to dynamic stall



Case: $\bar{\alpha} = 10^\circ$, $k = 0.05$, $\Delta\alpha = 5^\circ$

Beddoes-Leishman dynamic stall model



The Beddoes-Leishman dynamic stall model is composed of four algorithms:

- Unsteady attached flow module 5 $\rightarrow$ 1
- Non-linear trailing-edge separation module 1 $\rightarrow$ 2
- Leading-edge flow separation module 2 $\rightarrow$ 3
- Vortex shedding and lift module 3 $\rightarrow$ 4
- (Again) Non-linear trailing-edge separation module 4 $\rightarrow$ 5

Figure: Steady and unsteady polars for an abstract airfoil in pitching motion $\alpha = 14^\circ + 10^\circ \sin(\omega t)$ for $k = 0.053$

Beddoes-Leishman dynamic stall model

- The Beddoes-Leishman dynamic stall model is composed of four algorithms

Unsteady attached flow module
(quasi-steady airfoil aerodynamics)

Non-linear trailing-edge separation module

Leading-edge flow separation module

Vortex shedding and lift module

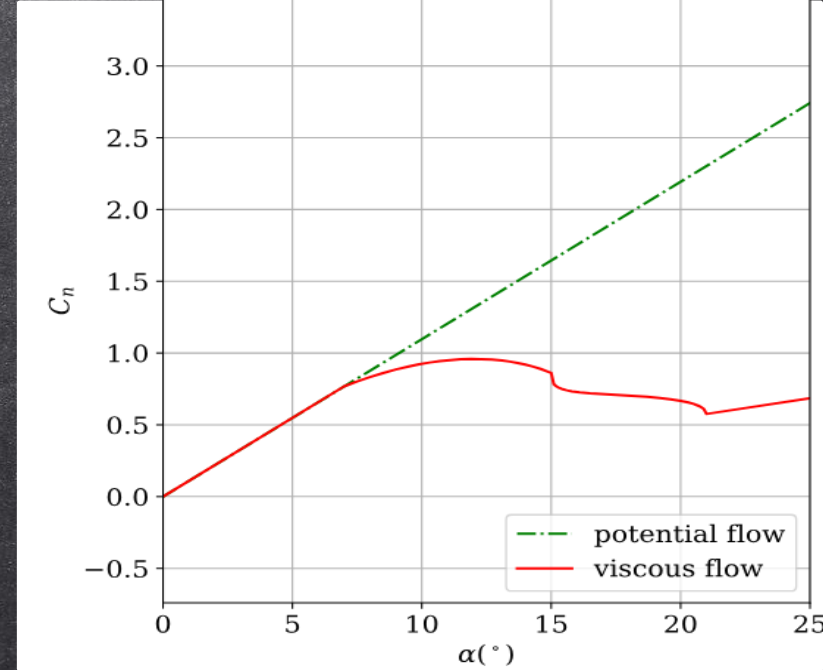
Non-linear trailing-edge separation

In BL model, the TE flow separation is modelled using the Kirchhoff and Helmholtz Theory that expresses the separated normal force as a function of the attached lift slope, AoA and a function of a pseudo-location of the separation point over the airfoil section f , which varies from $f=1$ at TE to $f=0$ at the LE.

$$C_{n_{st}} = \frac{dC_l}{d\alpha} \left(\frac{1 + \sqrt{f}}{2} \right)^2 (\alpha - \alpha_0)$$

f : steady experimental data or viscous simulations, $f(\alpha)$.

In unsteady flow, the separation experiences a lag due to the leading-edge pressure lag and boundary-layer development lag. These two sources of lag are modelled in two consecutive steps.



Potential flow and viscous flow normal load coefficient (C_n) in steady flow

Leading-edge pressure lag

The leading-edge pressure lag aims at determining an equivalent α of a lagged loading that produces a similar leading-edge pressure. The first step is to calculate a lagged potential flow load C_{np} , and then determine the equivalent α_f . The following equations are solved, including the integration of the deficit function D_{pf} .

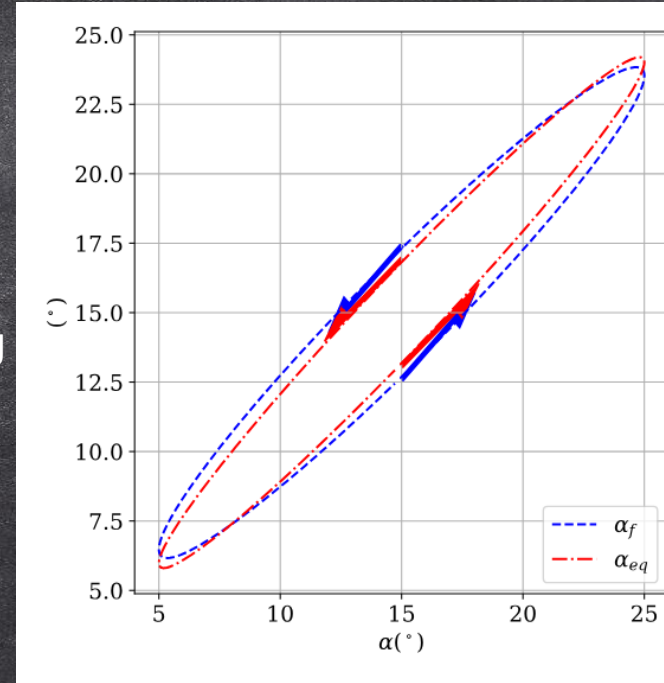
$$C_{np,} = C_{np} - D_{pf}$$

$$D_{pf}(s) = D_{pf(s-\Delta s)} e^{\frac{-\Delta s}{T_p}} + (C_{np(s)} - C_{np(s-\Delta s)}) e^{\frac{-\Delta s}{2T_p}}$$

$$\alpha_f = \frac{C_{np,}}{\frac{dC_l}{d\alpha}} + \alpha_0$$

$$f_p = f(\alpha_f)$$

T_p is mostly dependent on Mach number and in small amount in airfoil shape. $T_p = 1.7$ is used.



delay function for separation point
for boundary layer lag effects

Boundary-layer development lag

The boundary-layer development lag aims at calculating a second separation position function f_{bl} dependent on a boundary-layer development deficit function D_{bl}

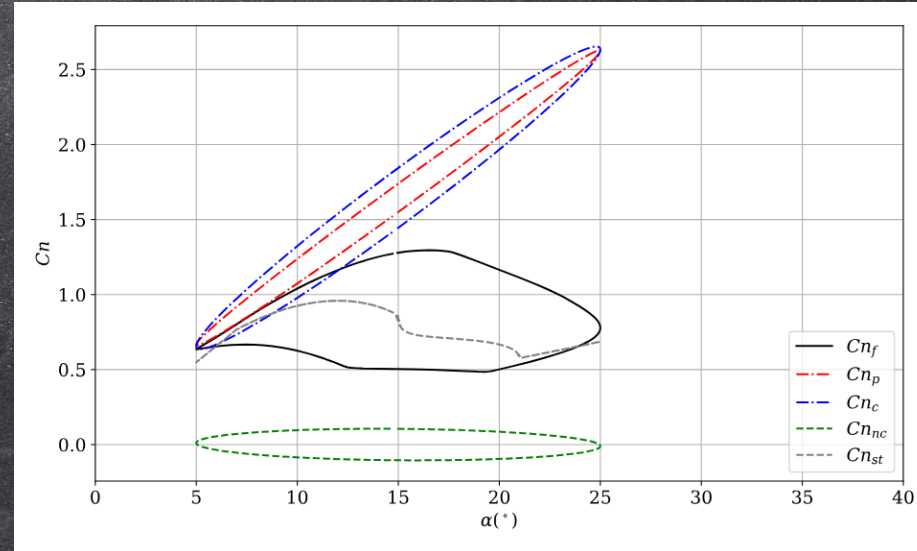
$$f_{bl} = f_p - D_{bl}$$

$$D_{bl(s)} = D_{bl(s-\Delta s)} e^{\frac{-\Delta s}{T_f}} + (f_{p(s)} - f_{p(s-\Delta s)}) e^{\frac{-\Delta s}{2T_f}}$$

T_f is dependent on airfoil shape. $T_f = 3.0$ is used.

The unsteady non-linear normal force coefficient including trailing edge separation is given by

$$C_{n_f(s)} = \frac{dC_l}{d\alpha} \left(\frac{1 + \sqrt{f_{bl}}}{2} \right)^2 (\alpha_{E(s)} - \alpha_0) + C_{n_{nc}(s)}$$



C_n due to trailing edge boundary layer separation comparing it with the other terms

Leading-edge flow separation module

In steady flow, it is possible to establish a relation between the onset of leading edge stall and a leading edge pressure gradient. This leading edge pressure gradient can be associated to a certain value of normal load.

Once this leading edge vortex occurs, it convects over the airfoil surface at a fraction of U_∞ . If we assume constant speed, then the location over the airfoil and wake is equivalent to a reduced time τ_v . We define that τ_v starts its counting when α_E increases and C_{nf} crosses a certain critical value C_{n1} . τ_v is reset to zero in the next cycle when once α_E increases and $C_{nf} < C_{n1}$. If we assume that the vortex convects with $0.45 U_\infty$, then the counter τ_v can be expressed by the equation

$$\begin{aligned}\tau_{v(s)} &= \tau_{v(s-\Delta s)} + 0.45\Delta s && \text{if } C_{nf} > C_{n1} \\ \tau_{v(s)} &= 0 && \text{if } C_{nf} < C_{n1} \text{ and } \Delta\alpha_E > 0\end{aligned}$$

Vortex shedding module

The leading-edge vortex causes an augmentation of the normal as the vortex travels over the surface of the airfoil lift augments the lift (normal force coefficient), a process that remains also as the vortex is convected downstream. The impact of the vortex increases as it convects over the surface of the airfoil, but decreases as it convects over the wake. The contribution by the vortex lift is determined by two equations. First, we define the forcing term

$$C_{v(s)} = C_{n_p(s)} \left(1 - \left(\frac{1 + \sqrt{f_{bl}}}{2} \right)^2 \right)$$

The vortex lift is determined by marching in time the solution of the following equation:

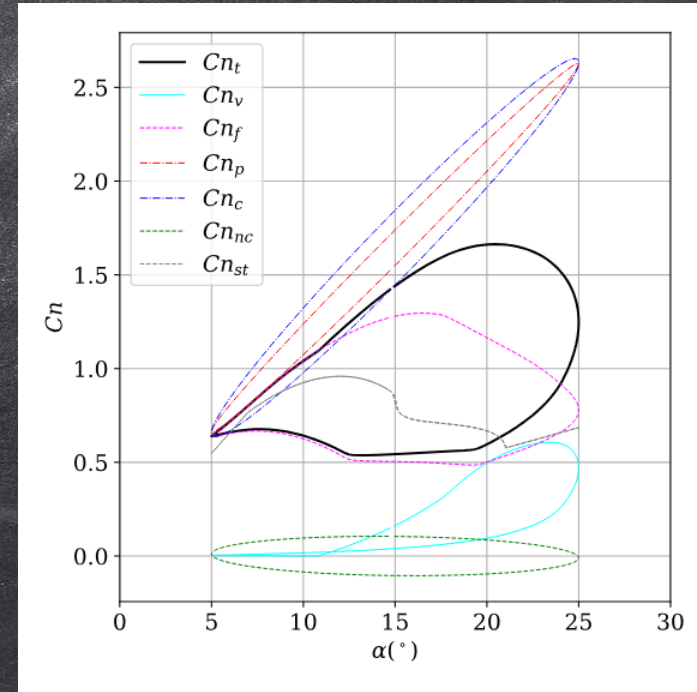
$$\begin{aligned} C_{n_v(s)} &= C_{n_v(s-\Delta s)} e^{\frac{-\Delta s}{T_v}} + (C_{v(s)} - C_{v(s-\Delta s)}) e^{\frac{-\Delta s}{2T_v}} & \text{if } 0 < \tau_v < T_{vl} \\ C_{n_v(s)} &= C_{n_v(s-\Delta s)} e^{\frac{-\Delta s}{T_v}} & \text{else} \end{aligned}$$

$T_v=6.0$ and $T_{vl}=5.0$ are used.

Vortex shedding module

The total unsteady non-linear normal force coefficient including trailing edge separation and leading-edge vortex-lift is given by

$$C_{n_t(s)} = C_{n_f(s)} + C_{n_v(s)} = \frac{dC_l}{d\alpha} \left(\frac{1 + \sqrt{f_{bl}}}{2} \right)^2 (\alpha_{E(s)} - \alpha_0) + C_{n_{nc}(s)} + C_{n_v(s)}$$



C_n including contribution by the leading edge vortex and the other terms.

In this module we have

- Derived the quasi-steady airfoil theory
- Learnt the basic application of models for unsteady airfoil aerodynamics using Duhamel Integral
- Learnt the physics of dynamic stall and the Beddoes-Leishman dynamic stall model

Rotor and Wake Aerodynamics

Unsteady panel method

Kelvin's Theorem: Rate of Change of the Circulation

Reminder

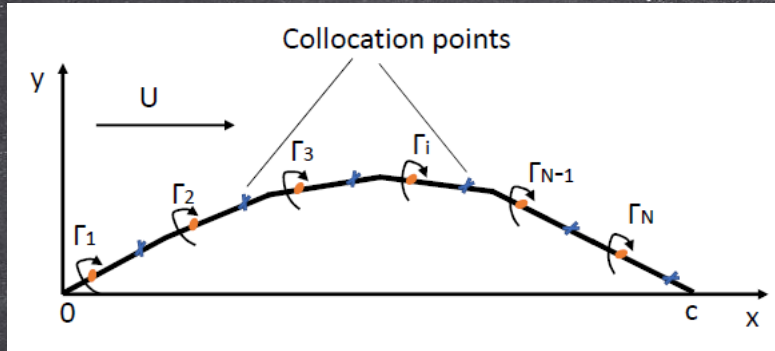
Implication of Kelvin's Theorem in an airfoil flow

The Bound Circulation on the Airfoil is the Symmetric of the Total Circulation in the Wake



$$\frac{D\Gamma}{Dt} = \frac{\Gamma_{airfoil} + \Gamma_{wake}}{\Delta t} = 0$$

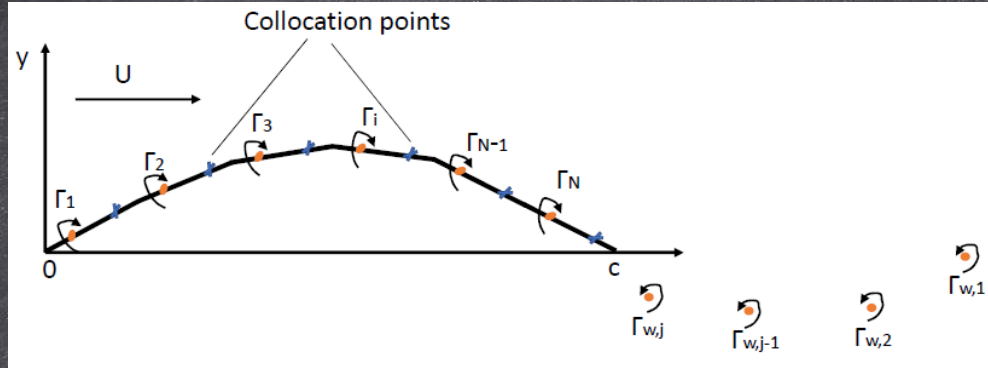
Steady panel method



N non-entry boundary condition eqs :

$$\sum_{k=1}^N a_{i,k} \Gamma_k + \vec{U} \cdot \vec{n}_i = 0, \text{ for } i = 1 \text{ to } N$$

Unsteady panel method



N non-entry boundary condition eqs :

$$\sum_{k=1}^N a_{i,k} \Gamma + \sum_{j=1}^j a_{i,j} \Gamma_{w,j} + \vec{U} \cdot \vec{n}_i = 0, \text{ for } i = 1 \text{ to } N$$

Helmholz's theorem: At jth time step, the strength of the newly shed vortex equals to the change of bound vorticity

$$\Gamma_{w,j} = -\Delta\Gamma = -\left(\sum_{i=1}^N \Gamma_i \Big|_{\text{time}=j} - \sum_{i=1}^N \Gamma_i \Big|_{\text{time}=j-1} \right)$$